

Normalization

Task 1. Normal Form Violations

Given table: `Booking_receipt(booking_id, passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, seat_numbers, ticket_price)`.

- **1NF Violation:** The `seat_numbers` attribute contains comma-separated values (e.g., "12A, 12B"), which violates the requirement of data atomicity. Every attribute in a relational database must hold only a single, atomic value.
- **2NF Violation:** Under a composite key (`booking_id, seat_number`), non-key flight-related attributes (`flight_number, departure_airport_name, arrival_airport_name, airline_name`) depend only on a subset of the candidate key (`booking_id`), introducing partial dependencies.
- **3NF Violation:** Transitive functional dependencies exist among non-key attributes: `flight_number → departure_airport_name → departure_city` and `flight_number → arrival_airport_name → arrival_city`. The city is determined by the airport rather than the primary key. Additionally, `passenger_passport_number → passenger_full_name`.

Task 2 & 3. Functional Dependencies and Step-by-Step Normalization

Functional Dependencies (FDs):

- `booking_id → flight_number`
- `flight_number → departure_airport_name, arrival_airport_name, airline_name`
- `departure_airport_name → departure_city`
- `arrival_airport_name → arrival_city`
- `passenger_passport_number → passenger_full_name`
- `(booking_id, seat_number) → passenger_passport_number, ticket_price`

Step 1: 1NF

The non-atomic `seat_numbers` column is decomposed so that each record represents a single seat. The primary key becomes composite:

- `Booking_receipt_1NF(booking_id, seat_number, passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, ticket_price)`

Step 2: 2NF

Partial dependencies are removed by isolating reservation/flight details from seat-specific data:

1. `Booking_2NF(booking_id, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name)`
2. `Ticket_Seat_2NF(booking_id [FK], seat_number, passenger_passport_number, passenger_full_name, ticket_price)`

Step 3: 3NF

Transitive dependencies are resolved by separating airports, flights, and passenger records into independent relations.

Final 3NF Relations (Primary Keys underlined, Foreign Keys marked):

1. Airports(airport_name, city)
2. Flights(flight_number, departure_airport_name [FK], arrival_airport_name [FK], airline_name)

(FK: departure_airport_name, arrival_airport_name →
Airports.airport_name)

3. Passengers(passenger_passport_number, passenger_full_name)
4. Bookings(booking_id, flight_number [FK])

(FK: flight_number → Flights.flight_number)

5. Tickets(booking_id [FK], seat_number, passenger_passport_number [FK], ticket_price)

(FK: booking_id → Bookings.booking_id, passenger_passport_number →
Passengers.passenger_passport_number)

Task 4. Flights Table 3NF Verification

- **Primary Key:** flight_id.
- **Functional Dependency:**

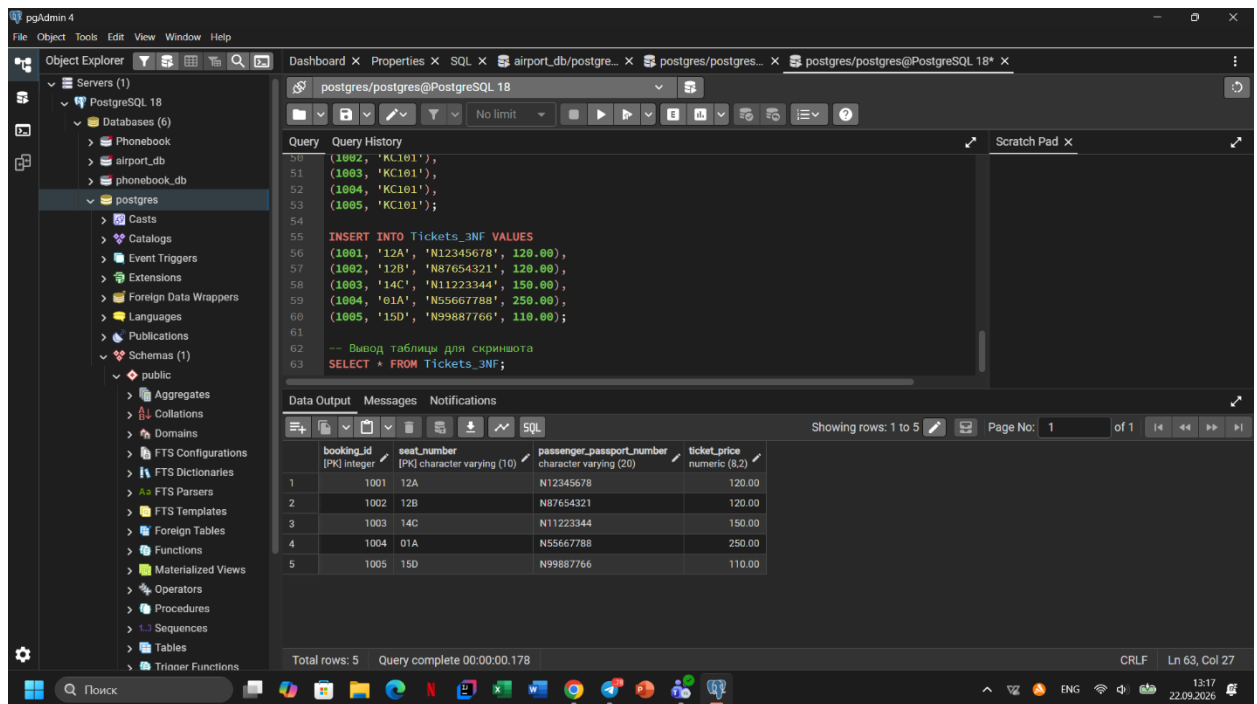
$\text{flight_id} \rightarrow \{\text{sch_departure_time}, \text{sch_arrival_time}, \text{departing_airport_id}, \text{arriving_airport_id}, \text{departing_gate}, \text{arriving_gate}, \text{airline_id}, \text{act_departure_time}, \text{act_arrival_time}, \text{created_at}, \text{updated_at}\}.$

- **Conclusion:** The `Flights` table is in **3NF**. All non-key attributes are directly and functionally dependent solely on the primary key `flight_id`. No transitive dependencies exist between non-key attributes because entity details are referenced via foreign key IDs (`departing_airport_id`, `arriving_airport_id`, `airline_id`) rather than storing non-key descriptive columns directly.

Task 5. 3NF Implementation and Sample Data

SQL

```
CREATE TABLE Airports_3NF (  
    airport_name VARCHAR(50) PRIMARY KEY,  
    city VARCHAR(50) NOT NULL  
);  
  
CREATE TABLE Flights_3NF (  
    flight_number VARCHAR(10) PRIMARY KEY,  
    departure_airport_name VARCHAR(50) NOT NULL REFERENCES  
Airports_3NF(airport_name),  
    arrival_airport_name VARCHAR(50) NOT NULL REFERENCES  
Airports_3NF(airport_name),  
    airline_name VARCHAR(50) NOT NULL  
);  
  
CREATE TABLE Passengers_3NF (  
    passenger_passport_number VARCHAR(20) PRIMARY KEY,  
    passenger_full_name VARCHAR(100) NOT NULL  
);  
  
CREATE TABLE Bookings_3NF (  
    booking_id INT PRIMARY KEY,  
    flight_number VARCHAR(10) NOT NULL REFERENCES Flights_3NF(flight_number)  
);  
  
CREATE TABLE Tickets_3NF (  
    booking_id INT NOT NULL REFERENCES Bookings_3NF(booking_id),  
    seat_number VARCHAR(10) NOT NULL,  
    passenger_passport_number VARCHAR(20) NOT NULL REFERENCES  
Passengers_3NF(passenger_passport_number),  
    ticket_price DECIMAL(8, 2) NOT NULL,  
    PRIMARY KEY (booking_id, seat_number)  
);  
  
INSERT INTO Airports_3NF VALUES  
( 'ALA', 'Almaty'),  
( 'NQZ', 'Astana');  
  
INSERT INTO Flights_3NF VALUES  
( 'KC101', 'ALA', 'NQZ', 'Air Astana');  
  
INSERT INTO Passengers_3NF VALUES  
( 'N12345678', 'Ivan Ivanov'),  
( 'N87654321', 'Anna Petrova'),  
( 'N11223344', 'John Smith'),  
( 'N55667788', 'Alikhan Omarov'),  
( 'N99887766', 'Elena Kim');  
  
INSERT INTO Bookings_3NF VALUES  
(1001, 'KC101'),  
(1002, 'KC101'),  
(1003, 'KC101'),  
(1004, 'KC101'),  
(1005, 'KC101');  
  
INSERT INTO Tickets_3NF VALUES  
(1001, '12A', 'N12345678', 120.00),  
(1002, '12B', 'N87654321', 120.00),  
(1003, '14C', 'N11223344', 150.00),  
(1004, '01A', 'N55667788', 250.00),  
(1005, '15D', 'N99887766', 110.00);
```



DDL Section

SQL

```

CREATE TABLE Airline_info (
    airline_id INT PRIMARY KEY,
    airline_code VARCHAR(30) NOT NULL,
    airline_name VARCHAR(50) NOT NULL,
    airline_country VARCHAR(50) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL,
    info VARCHAR(50) NOT NULL
);

```

```

CREATE TABLE Airport (
    airport_id INT PRIMARY KEY,
    airport_name VARCHAR(50) NOT NULL,
    country VARCHAR(50) NOT NULL,
    state VARCHAR(50) NOT NULL,
    city VARCHAR(50) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL
);

```

```

CREATE TABLE Passengers (
    passenger_id INT PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    date_of_birth DATE NOT NULL,
    gender VARCHAR(50) NOT NULL,
    country_of_citizenship VARCHAR(50) NOT NULL,
    country_of_residence VARCHAR(50) NOT NULL,
    passport_number VARCHAR(20) NOT NULL,

```

```

        created_at TIMESTAMP NOT NULL,
        updated_at TIMESTAMP NOT NULL
    );

CREATE TABLE Flights (
    flight_id INT PRIMARY KEY,
    sch_departure_time TIMESTAMP NOT NULL,
    sch_arrival_time TIMESTAMP NOT NULL,
    departing_airport_id INT NOT NULL REFERENCES Airport(airport_id),
    arriving_airport_id INT NOT NULL REFERENCES Airport(airport_id),
    departing_gate VARCHAR(50) NOT NULL,
    arriving_gate VARCHAR(50) NOT NULL,
    airline_id INT NOT NULL REFERENCES Airline_info(airline_id),
    act_departure_time TIMESTAMP NOT NULL,
    act_arrival_time TIMESTAMP NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL
);

CREATE TABLE Booking (
    booking_id INT PRIMARY KEY,
    flight_id INT NOT NULL,
    passenger_id INT NOT NULL REFERENCES Passengers(passenger_id),
    booking_platform VARCHAR(50) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL,
    status VARCHAR(50) NOT NULL,
    price DECIMAL(7,2) NOT NULL
);

CREATE TABLE Booking_flight (
    booking_flight_id INT PRIMARY KEY,
    booking_id INT NOT NULL REFERENCES Booking(booking_id),
    flight_id INT NOT NULL REFERENCES Flights(flight_id),
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL
);

CREATE TABLE Baggage_check (
    baggage_check_id INT PRIMARY KEY,
    check_result VARCHAR(50) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL,
    booking_id INT NOT NULL REFERENCES Booking(booking_id),
    passenger_id INT NOT NULL REFERENCES Passengers(passenger_id)
);

CREATE TABLE Baggage (
    baggage_id INT PRIMARY KEY,
    weight_in_kg DECIMAL(4,2) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL,
    booking_id INT NOT NULL REFERENCES Booking(booking_id)
);

CREATE TABLE Boarding_pass (
    boarding_pass_id INT PRIMARY KEY,
    booking_id INT NOT NULL REFERENCES Booking(booking_id),
    seat VARCHAR(50) NOT NULL,
    boarding_time TIMESTAMP NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL
);

```

```
ALTER TABLE Airline_info RENAME TO airline;
ALTER TABLE Booking RENAME COLUMN price TO ticket_price;
ALTER TABLE Flights ALTER COLUMN departing_gate TYPE TEXT;
ALTER TABLE airline DROP COLUMN info;
```

